

Documentation for BigQuery Fintech Dataset

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1 Preliminary Analysis of the Dataset

For this project, we make use of the BigQuery Fintech Dataset that is publicly available at <https://www.kaggle.com/dataset/fintech-dataset/data?select=loan.csv>. We first analyse the `loans.csv` dataset.

1.1 `loans.csv` Dataset

The dataset consists of 270299 rows and 18 columns occupying space of 172MB. A following is the list of the columns together with their datatypes:

Column	Data Type	Description
<code>loan_id</code>	Identifier	A digit from 1 to 270299 that represents the unique identifier for the loan.
<code>customer_id</code>	Identifier	A unique string identifier that identifies a customer.
<code>loan_status</code>	Categorical	Consists of 7 unique categories: Current, Fully Paid, Charged Off, Late (31-120 days), In Grace Period, Late (16-30 days), Default.
<code>loan_amount</code>	Numerical	The principal amount loaned.
<code>state</code>	Categorical	The resident US state of the borrower, corresponding to all 51 US states.
<code>funded_amount</code>	Numerical	The principal amount loaned.
<code>term</code>	Categorical (Binary)	The term taken to pay off the loan. A binary column of either 36 or 60 months.
<code>int_rate</code>	Numerical	Interest rate charged on the loan.
<code>installment</code>	Numerical	The amount of installment paid per month in the term.
<code>grade</code>	Categorical	Investment grade of the loan ranging from A-G.
<code>issue_d</code>	Date	The issue date of the loan.
<code>issue_date</code>	Date	The issue date of the loan.
<code>issue_year</code>	Categorical	The issue year of the loan, ranging from all years between 2012 and 2019 inclusive.
<code>pymnt_plan</code>	Boolean	Unsure. This however was a boolean column.
<code>type</code>	Categorical	The financing method of the loan. Consists of values like 'Individual', 'Joint' and 'DIRECT_PAY'.
<code>purpose</code>	Categorical	The reason for obtaining the loan.
<code>description</code>	Categorical	The reason for obtaining the loan.
<code>notes</code>	Degenerate	Consisted of the values 'desc' only.

Table 1: Column analysis of `loans.csv`. Dtypes are those inferred by `pandas.read_csv` (pandas 3.0.3, in which text columns take the dedicated `str` dtype). [†]`issue_d` and `issue_date` are stored as text (Jun-13 and June 2013 respectively) and are read as `str`; the dtype shown is the result of parsing them with `pandas.to_datetime`.

Column	Column Data Type
loan_status	numerical
loan_amount	numerical
state	categorical
term	categorical (binary)
int_rate	numerical
installment	numerical
grade	categorical
issue_d	date
type	categorical
purpose	categorical

Missing Values

The dataframe was free of missing values including placeholders for all columns except `description`.

Duplicate Columns

Upon inspection of the columns, it is noted that the following pairs of columns are duplicated: `loan_amount`, `funded_amount` for the principal amount, `issue_date`, `issue_d` for the date of issue, `purpose`, `description` for the purpose of loan.

Constant Columns

Furthermore, the `pymnt_plan` nearly constant and consisting of 99.99% of `False` variables.

There were also type issues noted in the `type` column, entries like 'individual' and 'INDIVIDUAL' counted as unique entries in pandas and were subsequently merged into single entries.

To handle all the data quality issues identified, a new dataset called 'loan_cleaned.csv' was created by dropping the duplicate columns (`funded_amount`, `issue_d`, `description`), the non-varying columns (`notes`, `pymnt_plan`) and the identifier columns (`loan_id`, `customer_id`).¹, and fixing the entry inconsistencies seen in the `type` column.

The final selected columns together with their data category types are given:

1.2 customer.csv dataset

The `customer.csv` dataset contains more personal information on the customers within the dataset. The dataset has 270299 rows and 11 columns occupying space of 132.4MB. A following is the list of the columns together with their datatypes:

Missing Values. The following columns possessed missing values:

1. `annual_inc_joint`. 93.05% missing values.
2. `emp_title`. 8.75% missing values.
3. `emp_length`. 6.93% missing values.

It might be a point to recheck the `addr_state`, `zip_code`, and `emp_title` columns for further missing values, because the column categories were very dense and quirky values such as 'n/a' in string format could be indicators of missing information that will be useful. But for now we proceed with

Assumption 1.1. `addr_state`, `zip_code`, and `emp_title` are all free of missing values.

¹This step will be reversed in the future to allow for joining with the other dataset `customer.csv`.

Column	Data Type	Description
customer_id	Identifier	A unique, hashed byte-string identifier for a customer (270299 unique values, one per row) - the intended join key with <code>loan.csv</code> .
emp_title	Categorical (high-cardinality)	Free-text job title of the borrower. 88932 unique values; 8.75% missing.
emp_length	Categorical	Length of employment, e.g. '1 year', '10+ years'. 11 unique values; 6.93% missing.
home_ownership	Categorical	Housing status of the borrower: RENT, MORTGAGE, OWN, ANY, OTHER, or NONE.
annual_inc	Numerical	Self-reported annual income of the borrower.
annual_inc_joint	Numerical	Self-reported joint annual income, only present for joint applications; 93.05% missing.
verification_status	Categorical	Whether income was verified: Verified, Source Verified, or Not Verified.
zip_code	Categorical	The first three digits of the borrower's zip code, e.g. '010xx'. 887 unique values.
addr_state	Categorical	The resident US state of the borrower, corresponding to all 51 US states.
avg_cur_bal	Numerical	Average current balance across the borrower's open credit lines.
Tot_cur_bal	Numerical	Total current balance across all of the borrower's credit lines.

Table 2: Column analysis of `customer.csv`. Dtypes and inferred semantic types are those produced by `loan.column_analysis.infer_semantic_type()`.